Real interpolation of variable Lorentz spaces

Removing the finite upper-exponent hypothesis

Automated Math Discovery Run `fa_banach_001`, lane 8

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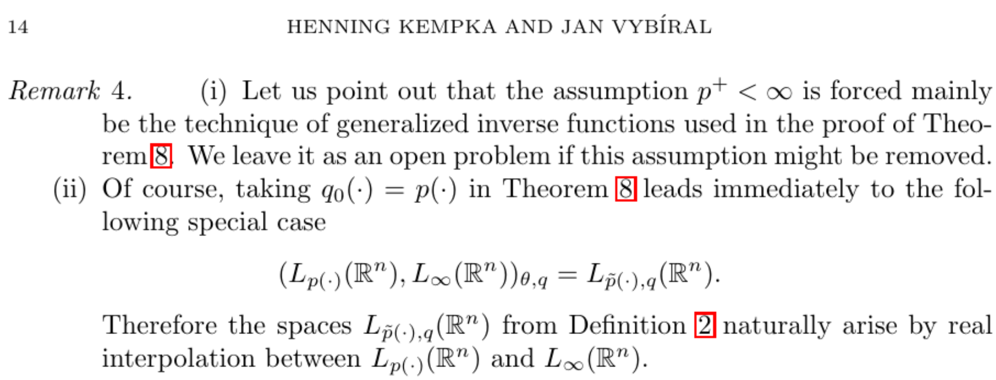
Result. The assumption $p^+ < \infty$ can be removed from Kempka–Vybíral's real-interpolation theorem. The formula remains valid when p is unbounded or equals ∞ on a set of positive measure.

1 Source question and conclusion

Kempka and Vybíral introduce variable Lorentz spaces and prove, under $p^+ < \infty$, that

$$(L_{p(\cdot),q_0(\cdot)}(\mathbb{R}^n), L_\infty(\mathbb{R}^n))_{\theta,q} \cong L_{\widetilde{p}(\cdot),q}(\mathbb{R}^n), \qquad \frac{1}{\widetilde{p}(\cdot)} = \frac{1-\theta}{p(\cdot)}.$$

Their Remark 4(i) says that finiteness of p^+ is used mainly for a generalized-inverse argument and asks whether it can be removed [1, Theorem 8 and Remark 4(i), PDF p. 14].



14 HENNING KEMPKA AND JAN VYBÍRAL

Remark 4. (i) Let us point out that the assumption $p^+ < \infty$ is forced mainly be the technique of generalized inverse functions used in the proof of Theorem 8. We leave it as an open problem if this assumption might be removed.

(ii) Of course, taking $q_0(\cdot) = p(\cdot)$ in Theorem 8 leads immediately to the following special case

$$(L_{p(\cdot)}(\mathbb{R}^n), L_\infty(\mathbb{R}^n))_{\theta,q} = L_{\tilde{p}(\cdot),q}(\mathbb{R}^n).$$

Therefore the spaces $L_{\tilde{p}(\cdot),q}(\mathbb{R}^n)$ from Definition 2 naturally arise by real interpolation between $L_{p(\cdot)}(\mathbb{R}^n)$ and $L_\infty(\mathbb{R}^n)$.

Figure 1: The open problem in arXiv:1210.1738, PDF page 14.

Let $\mathcal{P}$ denote, exactly as in [1], the measurable exponents from $\mathbb{R}^n$ into $(0, \infty]$ that are bounded away from zero. As usual, $L_{p(\cdot),q_0(\cdot)}$ is the corresponding variable Lorentz space, and $(\cdot,\cdot)_{\theta,q}$ is the real K-interpolation functor for quasi-Banach couples.

2 Proof intuition

For

$$h_k = \left\| \mathbf{1}_{\{|f|>2^k\}} \right\|_{L_{p(\cdot)}},$$

the target Lorentz quasi-norm is the ℓ_q norm of $2^k h_k^{1-\theta}$. This remains true on the set where $p = \infty$: the characteristic-function identity

$$\|\mathbf{1}_E\|_{L_{\widetilde{p}(\cdot)}} = \|\mathbf{1}_E\|_{L_{p(\cdot)}}^{1-\theta}$$

is exact.

The source proof encoded h through a generalized inverse, whose endpoint continuity led to $p^+ < \infty$. We instead work entirely on dyadic levels. At the weak Lorentz endpoint, any decomposition of f either leaves a large tail or has a large bounded part; this gives

$$K(f,t) \gtrsim \sup_k 2^k \min\{h_k, t\}.$$

At a constant Lorentz endpoint, clipping f at height 2^m gives the reverse estimate in terms of a tail of $(2^k h_k)$. The resulting tail operator is a one-sided discrete Hardy operator with a geometric kernel and is bounded on every ℓ_q, including $q < 1$. Monotonicity in the second Lorentz exponent then handles variable q_0.

Theorem 1. *Let $p, q_0 \in \mathcal{P}$, with no assumption on p^+ or q_0^+. Let $0 < \theta < 1$, $0 < q \leq \infty$, and define*

$$\frac{1}{\widetilde{p}(x)} = \frac{1-\theta}{p(x)},$$

with the convention $\widetilde{p}(x) = \infty$ when $p(x) = \infty$. Then

$$(L_{p(\cdot),q_0(\cdot)}(\mathbb{R}^n), L_\infty(\mathbb{R}^n))_{\theta,q} \cong L_{\widetilde{p}(\cdot),q}(\mathbb{R}^n)$$

with equivalent quasi-norms. Thus the answer to Remark 4(i) is affirmative.

3 Dyadic preliminaries

For a constant $0 < r \leq \infty$, the discretization from [1] gives

$$\|f\|_{L_{p(\cdot),r}} \asymp \|(2^k h_k)_{k\in\mathbb{Z}}\|_{\ell_r}. \tag{1}$$

No upper bound on p is needed here. Moreover, from the Luxemburg definition, for every measurable E,

$$\|\mathbf{1}_E\|_{L_{\widetilde{p}(\cdot)}} = \|\mathbf{1}_E\|_{L_{p(\cdot)}}^{1-\theta}. \tag{2}$$

Indeed, substituting $\mu = \lambda^{1/(1-\theta)}$ in the defining modular turns the condition for $L_{\widetilde{p}(\cdot)}$ exactly into the condition for $L_{p(\cdot)}$. This also covers the L_∞ part of the modular. Consequently,

$$\|f\|_{L_{\widetilde{p}(\cdot),q}} \asymp \|(c_k)_k\|_{\ell_q}, \qquad c_k := 2^k h_k^{1-\theta}. \tag{3}$$

Put $W = L_{p(\cdot),\infty}$ and, for $0 < r \leq \infty$, $R_r = L_{p(\cdot),r}$. We write $K_W(f,t) = K(f,t; W, L_\infty)$ and similarly for R_r.

Lemma 2 (two K-functional estimates)**.** *Uniformly for $t > 0$,*

$$K_W(f,t) \gtrsim \sup_{k\in\mathbb{Z}} 2^k \min\{h_k, t\}. \tag{4}$$

For $0 < r < \infty$, define

$$A_m = \left(\sum_{n \geq m} (2^n h_n)^r \right)^{1/r},$$

and for $r = \infty$ define $A_m = \sup_{n \geq m} 2^n h_n$. Then

$$K_{R_r}(f,t) \lesssim \inf_{m \in \mathbb{Z}} (A_m + t2^m). \tag{5}$$

Proof. Consider a decomposition $f = a + b$. If $\|b\|_\infty \leq 2^{k-1}$, then $|a| > 2^{k-1}$ on $\{|f| > 2^k\}$, whence $\|a\|_W \gtrsim 2^k h_k$. If $\|b\|_\infty > 2^{k-1}$, then $t\|b\|_\infty \gtrsim t2^k$. Taking the infimum over decompositions proves (4).

For (5), clip f radially at height 2^m: let

$$b(x) = \begin{cases} f(x), & |f(x)| \leq 2^m, \\ 2^m f(x)/|f(x)|, & |f(x)| > 2^m, \end{cases} \qquad a = f - b.$$

Then $\|b\|_\infty \leq 2^m$. If $k < m$, then $\{|a| > 2^k\} \subset \{|f| > 2^m\}$, while if $k \geq m$, then $\{|a| > 2^k\} \subset \{|f| > 2^k\}$. Thus the part below 2^m is bounded by the geometric sum with terms $2^k h_m$, and the part at and above 2^m is bounded by the corresponding tail of $(2^k h_k)$. Formula (1) gives $\|a\|_{R_r} \lesssim A_m$ for every $0 < r \leq \infty$. This proves (5). $\square$

4 The weak endpoint gives one inclusion

Lemma 3. *For every $0 < q \leq \infty$,*

$$(W, L_\infty)_{\theta,q} \hookrightarrow L_{\widetilde{p}(\cdot),q}.$$

Proof. Assume first $q < \infty$. If $h_k = \infty$ for some k, then (4) makes the interpolation quasi-norm infinite, as is the target quasi-norm, so there is nothing to prove. Otherwise partition the indices k with $0 < h_k < \infty$ according to

$$I_j = \{k : 2^j \leq h_k < 2^{j+1}\}.$$

Fix N and replace I_j temporarily by $I_{j,N} = I_j \cap \{-N, \ldots, N\}$. Whenever this finite set is nonempty, let $k_{j,N} = \max I_{j,N}$. Since h_k is nonincreasing in k,

$$\sum_{k \in I_{j,N}} c_k^q \lesssim 2^{k_{j,N} q} 2^{j(1-\theta)q}. \tag{6}$$

For $2^{j-1} \leq t \leq 2^j$, we have $t \leq h_{k_{j,N}}$, so Lemma 2 yields $K_W(f,t) \gtrsim 2^{k_{j,N}} t$. Hence

$$\int_{2^{j-1}}^{2^j} [t^{-\theta} K_W(f,t)]^q \, \frac{dt}{t} \gtrsim 2^{k_{j,N} q} 2^{j(1-\theta)q}.$$

The intervals in t are disjoint. Summing in j, using (6), and then letting $N \to \infty$ proves the claim by (3). For $q = \infty$, an infinite h_k is handled as above; otherwise choose j with $2^j \leq h_k < 2^{j+1}$ and evaluate (4) at $t = 2^j$. Taking the supremum in k gives the same conclusion. $\square$

5 Constant Lorentz endpoints give the reverse inclusion

Lemma 4. *For every* $0 < r \le \infty$ *and* $0 < q \le \infty$,

$$L_{\widetilde{p}(\cdot),q} \hookrightarrow (R_r, L_\infty)_{\theta,q}.$$

Proof. Let $D_m = 2^{-m}A_m$. The sequence D_m decreases at least geometrically: $D_{m-1} \ge 2D_m$. For nonzero f with $(c_k) \in \ell_q$, it has limits $D_m \to \infty$ as $m \to -\infty$ and $D_m \to 0$ as $m \to \infty$; the formulas below also give this directly. If $D_m \le t < D_{m-1}$, then (5) gives

$$K_{R_r}(f,t) \lesssim t2^m.$$

Therefore, for $q < \infty$,

$$\int_0^\infty [t^{-\theta}K_{R_r}(f,t)]^q \frac{dt}{t} \lesssim \sum_m 2^{mq} D_{m-1}^{(1-\theta)q}. \tag{7}$$

Suppose first $r < \infty$ and put $s = r/(1-\theta)$. A direct substitution of $c_n = 2^n h_n^{1-\theta}$ gives

$$2^m D_m^{1-\theta} = \left(\sum_{j\ge 0} 2^{-jr\theta/(1-\theta)} c_{m+j}^s\right)^{1/s}. \tag{8}$$

The operator on the right is bounded on ℓ_q for every $0 < q < \infty$. Indeed, if $q/s \le 1$, raise to the power q and use subadditivity; if $q/s \ge 1$, apply Young's convolution inequality to $(c_m^s)_m$. The geometric kernel belongs to every required ℓ_u. It is also bounded on ℓ_∞.

If $r = \infty$, the corresponding identity is

$$2^m D_m^{1-\theta} = \sup_{j\ge 0} 2^{-j\theta} c_{m+j}. \tag{9}$$

For finite q, its qth power is bounded by $\sum_{j\ge 0} 2^{-j\theta q} c_{m+j}^q$; summing in m proves ℓ_q boundedness. The ℓ_∞ case is immediate.

Equations (7)–(9), with an harmless shift from m to $m-1$, prove the assertion. When $q = \infty$, replace (7) by

$$\sup_{t>0} t^{-\theta} K_{R_r}(f,t) \lesssim \sup_m 2^m D_{m-1}^{1-\theta}$$

and use the same sequence estimates. $\square$

6 Proof of the theorem

Let $r = q_0^- \in (0,\infty]$. The monotonicity theorem for variable Lorentz spaces [1, Theorem 7(i)] gives

$$R_r = L_{p(\cdot),r} \hookrightarrow X := L_{p(\cdot),q_0(\cdot)} \hookrightarrow W = L_{p(\cdot),\infty}.$$

Monotonicity of the K-functional consequently gives

$$(R_r, L_\infty)_{\theta,q} \hookrightarrow (X, L_\infty)_{\theta,q} \hookrightarrow (W, L_\infty)_{\theta,q}.$$

Lemma 4 embeds the target space into the left member, while Lemma 3 embeds the right member into the target. This proves Theorem 1.

Notice where the endpoint and quasi-Banach cases enter. The set $\{p = \infty\}$ is covered by the exact identity (2). The proof of (4) uses only a level-set inclusion, not a triangle inequality, so it remains valid when $p^- < 1$. The geometric Hardy proof is valid for every $r > 0$, including $r = q_0^- < 1$. Finally, $q_0^- = \infty$ is covered by (9), and $q = \infty$ uses the displayed supremum estimates. No limiting argument in p^+ and no local convexity are used.

7 Verification and novelty notes

The main verifier targets are:

1. the direction of the two monotone Lorentz embeddings in the final sandwich;
2. the tail identity (8) and its exponents;
3. the endpoint $r = \infty$ formula (9);
4. the fact that all constants are independent of dyadic truncations.

The included script tests the finite versions of the two geometric Hardy operators over random positive sequences and parameters. These tests are only sanity checks; the proof above is analytic.

A bounded official-arXiv search through 11 August 2026 used the title and arXiv id 1210.1738 together with the phrases "real interpolation", "variable Lorentz", "$p^+ = \infty$", and "remove $p^+ < \infty$". It found papers on variable-exponent real interpolation and Hardy–Lorentz spaces, but no paper removing this hypothesis from Theorem 8. Recent papers on the different mixed sequence space $\ell^{q(\cdot)}(L^{p(\cdot)})$ do not answer this interpolation question. The result should therefore receive expert proof review and an independent MathSciNet/zbMATH novelty check before use.

References

[1] H. Kempka and J. Vybíral, *Lorentz spaces with variable exponents*, Math. Nachr. 287 (2014), 938–954; arXiv:1210.1738.